Indian Statistical Institute

M.Tech. (CS), Final of Second Semester Examination, 2024-25

Database Management Systems

Full Marks: 50 Date: 30-04-2025 Time: 2.5 Hours

Answer any *five* of the following questions

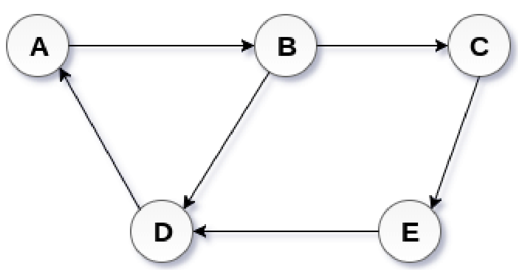
$5 \times 10 = 50$

1. (a) Suppose two publishers maintain their databases, say P1 and P2, following the schema given below.

P1TABLE1 = <*id* : *string*, *title* : *string*, *cost* : *integer*, *authors* : *string*, *Edition* : *integer*>
P2TABLE1 = <*ISBN* : *integer*, *title* : *string*, *cost* : *real*, *authors* : *string*>
P2TABLE2 = <*ISBN* : *integer*, *Edition* : *integer*, *Date* : *date*>

The tables under each database P*i* is separately numbered as P*i*TABLE*j*. Note that ISBN denotes the International Standard Book Number. Suggest a method of integrating both the databases into a single database.

(b) A directed graph can be saved in a relation in terms of its adjacency list denoted by the attributes <*source-vertex*, *target-vertex*>. For example, the following graph **G** (shown in left) can be represented as the relation **R** (shown in right).



source-vertex	target-vertex
A	B
B	C
B	D
C	E
E	D
D	A

G **R**

A triangle is defined as a directed cycle of length 3. Note that in the graph **G**, (A, B, D) is a triangle, whereas (A, B, C, E, D) is a cycle but not a triangle. Write a relational algebra expression to return all the triangles in any arbitrary **G** from its respective relation **R**.

(c) Can a discriminating attribute of a weak entity set be a foreign key in the same entity set? Justify your answer.

4+4+2

2. Let there be a relational schema *R* = <U, V, W, X, Y, Z> holding the set of functional dependencies FD = {U → V, VW → X, X → YZ, X → Z}.
(i) Find out the closure of the attribute set UW.
(ii) Derive all other non-trivial functional dependencies that can be obtained from FD.

(iii) Decompose R into any arbitrary subrelations that are lossless-join. Justify your answer.
(iv) Decompose R into any arbitrary subrelations that are dependency preserving. Justify your answer.

2+2+2+4

3. (a) Consider the relations $R1$ = <<u>A</u>, B, C> and $R2$ = <<u>B</u>, D, E> having 5040 and 24 tuples, respectively. Notably the primary keys in the respective relations are underlined therein. The number of distinct values of B in $R1$ is 7. The range of values of C and D in $R1$ and $R2$ are [50, 120] and [0, 60], respectively. Estimate the size of the following query considering the attribute values are uniformly distributed.

$$\sigma_{(C \leq 100) \wedge (B!=0)} (R1) \bowtie \sigma_{(D \geq 100)} (R2)$$

(b) Prove the following equivalence relation of relational algebra assuming θ is valid for R_1.

$$\sigma_\theta (R_1 - R_2) \equiv \sigma_\theta (R_1) - R_2$$

(c) Cite an example where both the sort merge join and hash join are equally good physical plans for performing natural join.

5+3+2

4. (a) Let there be a relation with data items A and B. Derive the *precedence graph* from the following schedule of transactions working on the said relation. Show that the given schedule is *conflict serializable*. Justify whether the given schedule is *view serializable* or not.

Transaction T_1	**Transaction T_2**	**Transaction T_3**	**Transaction T_4**
			read(A)
	read(A)		
	A ← A + 12		
		read(A)	
		A ← A * 0.1	
write(B)			
	write(A)		
		read(B)	
	write(B)		
	Commit		

(b) Show the steps to identify the serializability order from the precedence graph that you derived from the schedule given in (a).

(5+2)+3

5. Identify whether any problem will occur in executing each of the following schedules. If so, justify your answer. Note that commit statements are not explicitly mentioned within the schedules.

(i)

T_1	T_2	T_3	**Timestamp**
lock-X(A)			1
read(A)			2
A = A + 5			3
write(A)			4
	lock-S(B)		5
	read(B)		6
	lock-S(A)		7

unlock(A)			8
lock-S(B)			9
read(B)			10
B = B – 15			11
		lock-X(B)	12
write(B)			13
unlock(B)			14
		unlock(B)	15
	unlock(B)		16

(ii)

T_1	T_2	T_3	**Timestamp**
lock-X(B)			1
	lock-X(A)		2
	read(A)		3
	A = A + A*0.2		4
	write(A)		5
	unlock(A)		6
lock-X(A)			7
read(A)			8
A = A – 100			9
write(A)			10
unlock(A)			11
		lock-X(A)	12
		read(A)	13
		A = A + 10	14
		write(A)	15
		unlock(A)	

(iii)

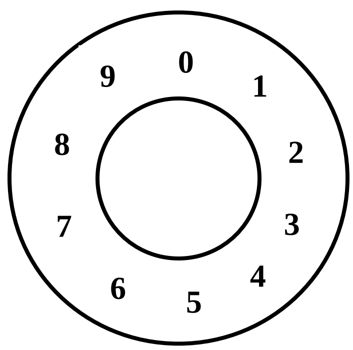
T_1	T_2	**Timestamp**
lock-S(B)		1
read(B)		2
B = B + 10		3
	lock-S(B)	4
	read(B)	5
	lock-S(A)	6
	read(A)	7
	A = A * 0.05	8
	write(A)	9
	unlock(A)	10
	unlock(B)	11
upgrade(B)		12
write(B)		13

6+3+1

6. (a) Convert the following collection of JSON documents into minimum number of relations (tables). Note that the relations may contain NULL values.

```
{
        "Roll" : 1,
        "Name" : "Shyam",
        "Centre" : "Kolkata",
        "DOB" : 30-04-2025,
}
{
        "Roll" : 1,
        "Name" : "Somaiya",
        "Centre" : "Bangalore",
        "Rank" : 5,
        "Grade" : 8.5,
}
{
        "Roll" : 1,
        "Name" : "Malik",
        "Centre" : "Delhi",
        "Rank" : 5,
        "Grade" : 8.5,
}
{
        "Roll" : 1,
        "Name" : "Stefan",
        "Centre" : "Chennai",
        "Rank" : 5,
}
```

(b) Let there be a distributed environment with 7 objects (labeled as 1 to 7) assigned to 4 different nodes (node 1, 2, 3 and 4) implementing consistent hashing. Let the objects are assigned positions on the hash table (as shown below) using the hash function *L*%10, where the object label is *L*. On the other side, the nodes are assigned positions using the hash function (*N**2)%10, where the node label is *N*. Note that a node and an object can both be hashed into the same index, thereby denoting that the object gets assigned to the respective node. Suppose a new object (labeled as 8) gets hashed into the environment after the replacement of one node (node 2 replaced with node 5). What will be the new assignment of objects to different nodes? Is this replacement better or worse in terms of load balancing?



(c) Why is consistency a mandatory requirement in both SQL and NoSQL databases but not the others like atomicity, isolation and durability?

4+4+2

7. Show appropriate examples of the following things.

(i) Data validation with consistency check.

(ii) A pair of tables that will return the same result for left and right outer joins.

(iii) Discriminating attribute of a weak entity set.

(iv) A table that satisfies Sixth Normal Form (6NF).

(v) Concurrency control with validation-based protocol.

2+2+2+2+2